



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

TRAFFIC AND POLLUTION CORRELATION

The domain of the Project:
POWER BI

Team Mentor (and their designation):
Prof. Abhishek Ramachandran
Finance & Analytics Professional

Team Member:
Shivom Bhattacharjee

Period of the project

October 2025 to December 2025



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Declaration

The project titled “Traffic and Pollution Correlation” has been mentored by Prof. Abhishek Ramachandran, organised by SURE Trust, from October 2025 to December 2025, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Member:

Mr. Shivom Bhattacharjee -- Signature

Prof. Abhishek Ramachandran
Finance & Analytics Professional
Dell Technologies

Prof. Radhakumari
Executive Director & Founder
SURE Trust



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Executive Summary

This project focuses on analyzing the relationship between **traffic congestion and air quality (AQI)** using an interactive **Power BI dashboard** designed for smart city decision-making. The primary objective was to understand how traffic patterns, weather conditions, and zonal characteristics influence pollution levels across the city and to identify areas requiring targeted intervention.

The analysis was conducted using multiple integrated datasets, including **Traffic, Pollution, Weather, Date, and Zone-level data**. Advanced **DAX calculations** were implemented to derive key indicators such as **Average AQI, Traffic Congestion %, Traffic Efficiency %, Hotspot Count, and AQI–Congestion Correlation**. The dashboard was structured into multiple interactive pages—Main, Traffic, Pollution, Hotspot, and Zone Drill-Through—allowing users to explore trends at both city and zone levels with seamless navigation and filtering.

Key findings reveal that while **average traffic congestion remains relatively stable across months, AQI shows significant fluctuations**, indicating that pollution levels are influenced by multiple factors beyond congestion alone. Certain zones repeatedly experience **high AQI and frequent hotspot days**, identifying them as persistent pollution hotspots. Seasonal analysis shows that **rainfall can temporarily reduce AQI in some zones**, but this effect is **not consistent across the city**, highlighting the limited reliability of weather as a standalone pollution control factor. Traffic efficiency metrics suggest that zones with **lower speeds and higher congestion exposure** tend to experience worse air quality outcomes.

Based on these insights, the project recommends **targeted traffic management strategies** in high-risk zones, including congestion control, improved traffic flow, and stricter monitoring during peak pollution periods. Continuous zone-level monitoring, combined with data-driven policy decisions, can help city authorities reduce pollution exposure, improve public health, and support long-term **smart city planning** initiatives.



Introduction

Background and Context

Rapid urbanization and increasing vehicle density have significantly impacted air quality in modern cities. Traffic congestion is often considered a major contributor to air pollution, especially in densely populated urban zones. With growing emphasis on **smart city initiatives**, there is a strong need for data-driven systems that can continuously monitor traffic behavior, pollution levels, and environmental factors to support effective planning and policy decisions.

This project was developed in this context to provide a **visual, analytical, and interactive solution** that helps stakeholders understand pollution patterns and their possible drivers at both city and zone levels.

Problem Statement and Goals

Problem Statement:

Urban authorities lack an integrated analytical view to understand how traffic congestion, weather conditions, and zone-level factors collectively affect air quality.

Project Goals:

- Analyze the relationship between **traffic congestion and AQI**
- Identify **seasonal pollution patterns**
- Detect **high-risk hotspot zones**
- Enable **zone-level analysis** through drill-through
- Provide **actionable insights** for smart city planning and pollution control

Scope and Limitations

Scope:

- Analysis based on traffic, pollution, weather, date, and zone datasets
- City-level and zone-level AQI analysis
- Seasonal and monthly trend evaluation
- Identification of pollution hotspots
- Interactive Power BI dashboard with KPIs, correlations, and drill-through

Limitations:

- Analysis is based on available historical data only
- Other pollution sources (industrial, construction, domestic emissions) are not included



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- Correlation does not imply direct causation
- Real-time sensor data integration is outside the project scope

Innovation Component

- Integration of **traffic, pollution, and weather datasets** into a single analytical platform
- Use of **interactive drill-through pages** for zone-specific insights
- Identification of **repeated hotspot zones** instead of one-time pollution spikes
- Visual correlation of **hotspot days with AQI severity**
- Dashboard designed to support **decision-making rather than just reporting**

This approach transforms remembered data into **actionable intelligence**, making the project suitable for **real-world smart city applications**.



Project Objectives

Objectives and Goals

- To quantify the relationship between **traffic congestion and AQI**
- To analyze **monthly and seasonal pollution trends**
- To identify **high-risk hotspot zones** across the city
- To assess the impact of **rainfall and temperature** on air quality
- To design an **interactive Power BI dashboard** for easy interpretation

Expected Outcomes and Deliverables

- A fully functional **Power BI dashboard** with multiple analytical pages
- Clear visualization of **traffic trends, pollution patterns, and hotspots**
- Zone-level drill-through analysis for deeper insights
- Actionable insights to support **smart city planning and pollution control**
- A scalable analytical framework that can be extended to real-time monitoring in the future



Methodology and Results

Methods/Technology Used

- Correlation Analysis: Correlation coefficient calculated to quantify the linear relationship between congestion (%) and AQI across months.
- Aggregation & Time-Series Analysis: Monthly and daily aggregations performed to identify seasonal patterns and temporal trends in both traffic and pollution.
- Hotspot Detection: Threshold-based approach (AQI > 150 for "hotspot" days, AQI > 200 for "critical alerts") to isolate high-risk periods and zones.
- Zone-Level Drill-Down: Granular filtering and comparison across 50 zones to understand localized pollution behavior and zone level analysis.
- Descriptive Statistics: Calculation of averages, counts, and percentages to summarize KPI values and enable quick executive-level assessment.

Tools/Software Used

- Power BI Desktop: Data modeling, DAX formula authoring, and interactive dashboard design.
- DAX (Data Analysis Expressions): Custom KPI calculations including AQI–Congestion Correlation %, Seasonal Pollution Index, Hotspot Count, and Repeat Hotspot Zones.
- Excel datasets – Source data for traffic, pollution, and weather.

Data Collection Approach

- Traffic Data: Congestion (%), Average Speed (km/h), collected daily across all zones with timestamp granularity.
- Pollution Data: Air Quality Index (AQI), PM2.5 levels, recorded daily per zone; readings synchronized with traffic timestamps.
- Weather Data: Rainfall (mm), Temperature (°C), collected at city level or zone level to correlate with pollution spikes.
- Zone Information: Geographic zone names and identifiers to enable spatial filtering and zone-wise drill-through analysis.
- Date Dimension: Enabled month, season, and year-level aggregation for trend analysis.

Project Architecture

Data Layer

- Four source tables: Traffic, Pollution, Weather, and Zones.
- A Date dimension table enabling time-based hierarchies (Year → Month → Day).
- Relationships established on Zone and Date as key link fields to support cross-filtering.



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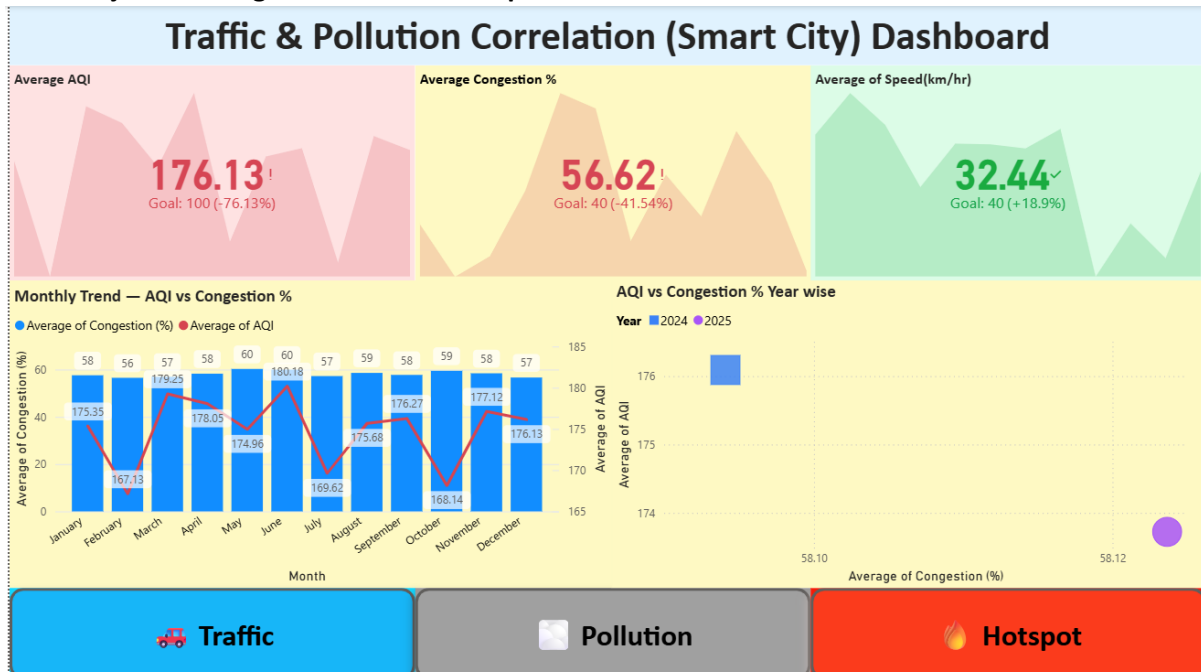
Analytics Layer (DAX Calculations)

- KPI Layer: Measures for Average AQI, Average Congestion, Traffic Efficiency, Seasonal Pollution Index.
- Correlation Layer: AQI–Congestion Correlation % calculated at overall and monthly granularity.
- Hotspot Layer: Pollution Alerts Triggered (>200 AQI), High-Risk Zones (>150 AQI), Repeat Hotspot Zones, Hotspot Intensity (avg AQI in hotspots).
- Trend Layer: Monthly trend measures and year-over-year comparisons.

Presentation Layer (5 Pages)

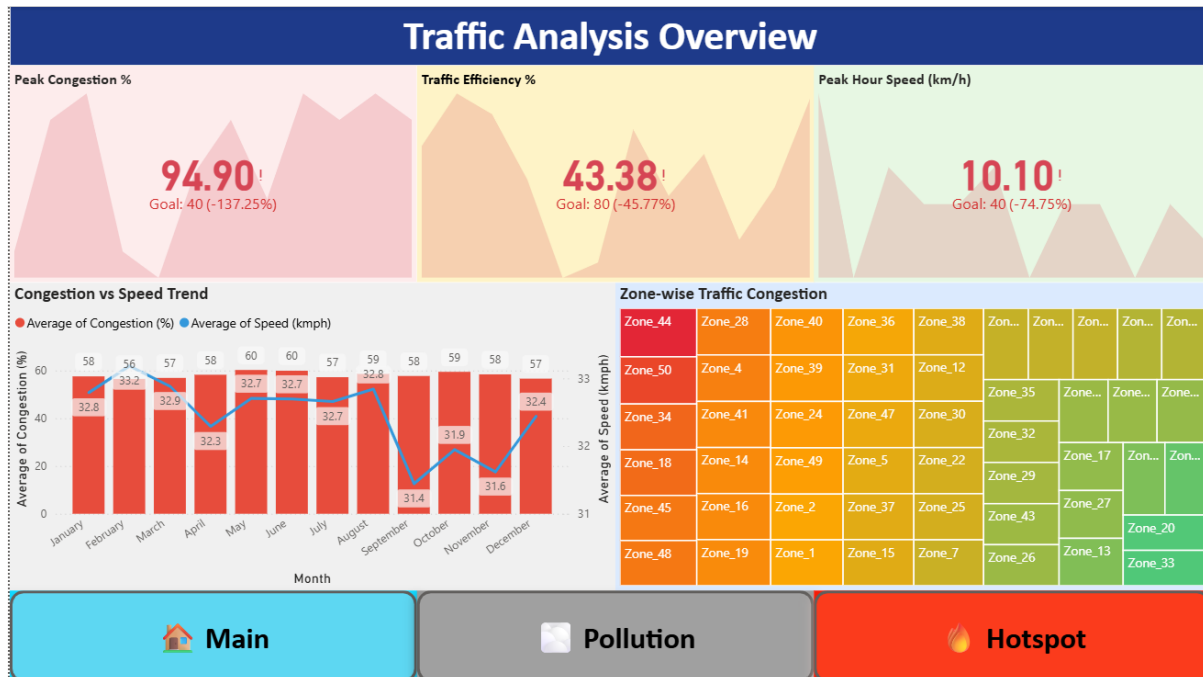
- Main Page: Overall KPIs and correlation chart showing citywide story.
- Traffic Page: Congestion trends, speed analysis, zone-wise traffic heatmap.
- Pollution Page: Seasonal pollution index, clean air days %, AQI distribution, rainfall correlation.
- Hotspot Page: Repeat hotspot zones, average AQI in hotspots, hotspot intensity gauge, zone-wise hotspot treemap.
- Zone Drill-Through: Detailed view per zone with hotspot days, peak AQI, rainfall, and monthly trends.

Final Project Working Screenshots and Explanation



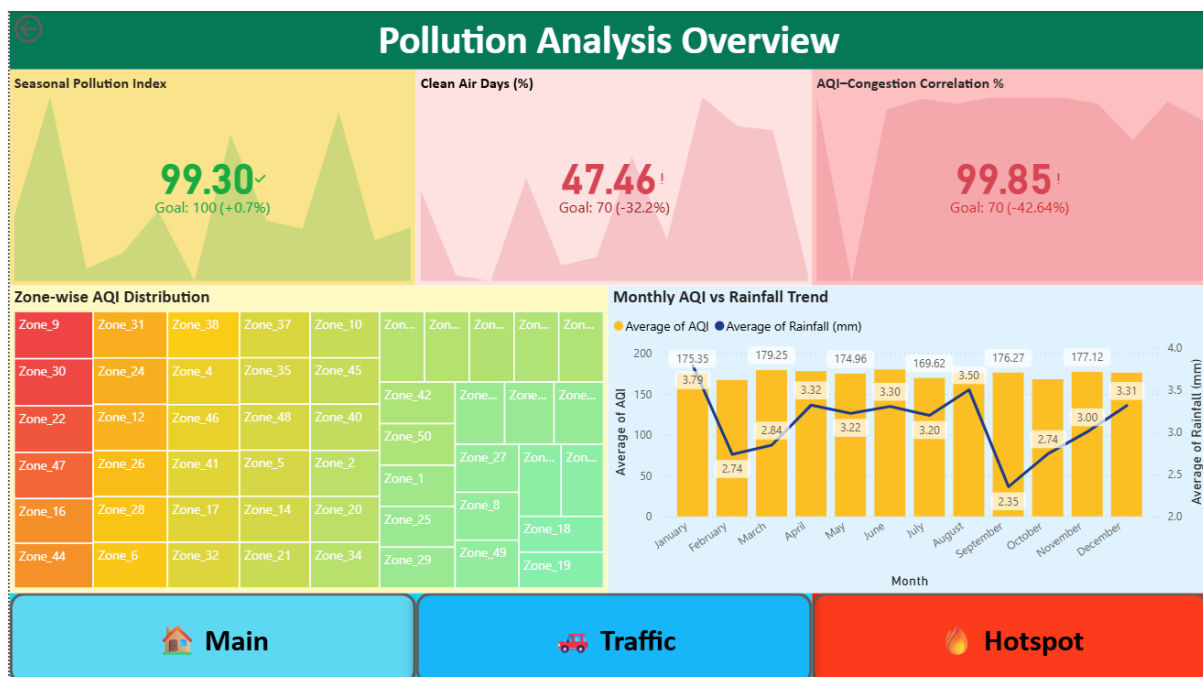
Main Page Overview:

The main page displays three headline KPIs: Average AQI, Average Congestion, Average Speed and a monthly trend chart showing AQI vs. Congestion %. The trend reveals that despite stable congestion, AQI fluctuates significantly, indicating seasonal drivers beyond traffic alone. A year-over-year scatter plot (right side) confirms the correlation pattern holds across 2024 and 2025 data.



Traffic Analysis Overview:

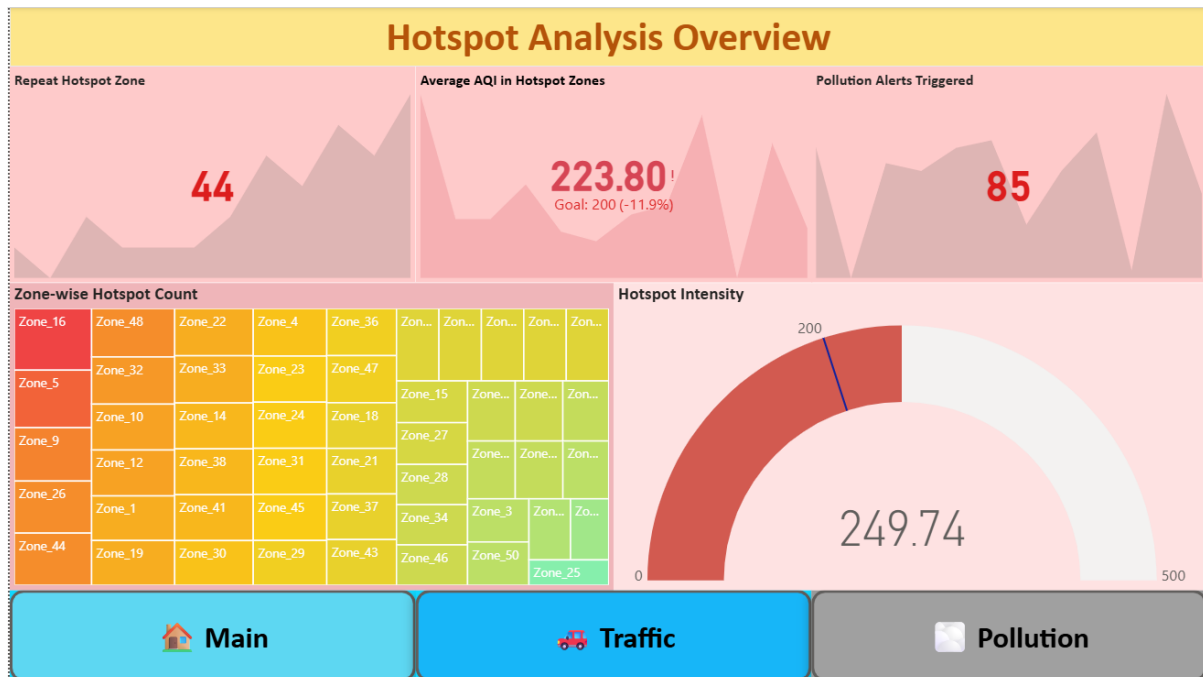
This page highlights three traffic KPIs: Peak Congestion, Traffic Efficiency, and Peak Hour Speed, all marked red to signal suboptimal performance. The "Congestion vs. Speed Trend" line chart shows inverse relationship—when congestion rises, average speed drops. The zone-wise traffic heatmap (right) reveals that Zone 44, Zone 50 and Zone 34 are the most congested areas.





Pollution Analysis Overview:

Three KPIs summarize pollution status: Seasonal Pollution Index, Clean Air Days, and AQI–Congestion Correlation. The zone-wise AQI distribution heatmap shows Zones 9, 30, 22, and 47 consistently in the red (high AQI), while a supporting chart plots monthly AQI against rainfall, revealing that rain does reduce pollution but the effect is inconsistent across zones.

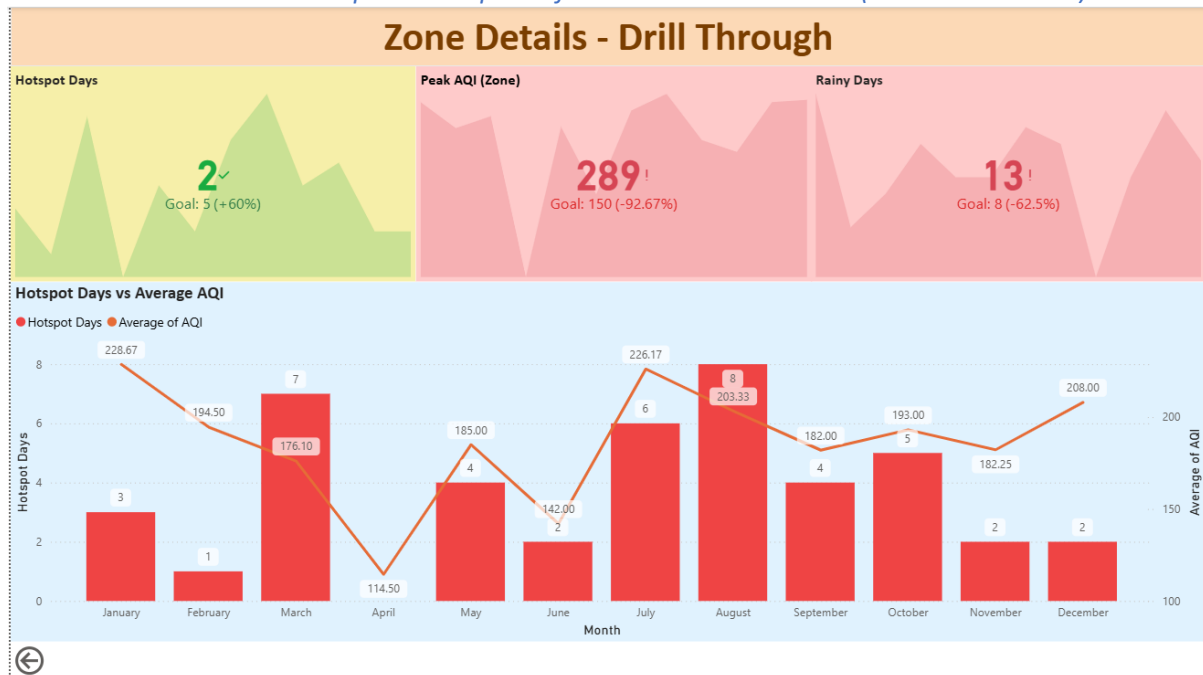


Hotspot Analysis Overview:

Top KPIs display Repeat Hotspot Zones, Average AQI in Hotspot Zones and Pollution Alerts Triggered. The zone-wise hotspot count treemap uses color intensity to show which zones trigger the most hotspot days—Zone 16, Zone 5, Zone 9. The Hotspot Intensity gauge (right) shows 249.74, well above the 200 target, indicating severe pollution in hotspot periods.



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Zone Details – Drill Through:

Clicking a zone from the hotspot page reveals zone-specific insights: Hotspot Days, Peak AQI, and Rainy Days. A combined bar–line chart shows monthly hotspot days and average AQI, revealing that August had the highest hotspot count (8 days) and peak AQI (203.33).

Project GitHub Link:

[sure-trust/SHIVOM-BHATTACHARJEE-g20-sql-and-powebi](https://github.com/sure-trust/SHIVOM-BHATTACHARJEE-g20-sql-and-powebi)



Learning and Reflection

New Learnings

- Learned how to design an end-to-end Power BI solution, from data modelling to interactive multi-page dashboards.
- Improved DAX skills by building advanced KPIs like AQI–Congestion Correlation %, Seasonal Pollution Index, Hotspot Days, and Traffic Efficiency.
- Understood how to interpret traffic, pollution, and weather data together to derive meaningful, real-world insights.
- Gained experience in structuring an executive-friendly story using KPIs, charts, and drill-through for decision-makers.
- Enhanced understanding of smart-city concepts such as hotspot detection, zone-level risk, and impact of congestion on AQI.

Overall Experience

- Working as a single-member team gave full ownership of problem understanding, solution design, and presentation.
- Managing both technical work (DAX, modelling, visuals) and communication (slides, report, recommendations) was challenging but rewarding.
- Building the hotspot and drill-through pages felt particularly impactful because they translate raw data into direct action points for authorities.
- The project increased confidence in handling real-world analytics problems end-to-end and presenting them to senior stakeholders.



Conclusion and Future Scope

Recap of Objectives and Achievements

- Analysed the relationship between traffic congestion and air quality using integrated traffic, pollution, weather, date, and zone datasets, and quantified it through KPIs such as AQI–Congestion Correlation %.
- Identified high-risk hotspot zones and repeat hotspot behaviour using metrics like hotspot days, pollution alerts, and average AQI in hotspot zones, enabling targeted interventions.
- Detected seasonal patterns in AQI, congestion, and rainfall to highlight months and periods with elevated pollution risk for citizens.
- Designed and implemented a multi-page interactive Power BI dashboard (Main, Traffic, Pollution, Hotspot, Zone Drill-through) to support decision-making for smart-city planning.
- Translated analytical insights into clear executive-level recommendations on prioritising zones, improving traffic efficiency, planning seasonal actions, and setting up continuous monitoring.

Future Scope of this Project

The current solution can be extended by incorporating more granular data, such as hourly traffic and AQI readings, to capture peak-time effects and short-term pollution spikes in greater detail. Additional data sources like industrial emission inventories, construction activity logs, or public transport usage can be integrated to separate the impact of non-traffic pollution contributors and refine the attribution of AQI changes. Predictive models and what-if simulations could be built on top of the existing KPIs to estimate how specific policy actions—such as odd–even schemes, truck restrictions, or signal optimisation—would affect future AQI levels. At a platform level, the dashboard can be connected to near real-time data feeds and alert systems so that authorities receive automatic notifications when zones cross critical thresholds, turning the project into an operational monitoring and control tool rather than just an analytical report.